Integrated Library Management System

Complete user, administrator and deployment manual

Version 2.0 | 23 October 2026

RBCET / RBMI Central Library
RBCP Library
RBCN Library
RBGN Library

Live: https://rbmi-ilms.vercel.app
Administrator: rahul.r@rbmi.in

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01 / Start here and library directory

Open https://rbmi-ilms.vercel.app in a current browser. RBMI ILMS 2.0 uses React, shared Supabase PostgreSQL and Supabase email/password authentication. GitHub and hosting dashboard accounts are separate from library sign-in accounts.

How to use this section

1. Sign in with your registered email and private password. Verify that the connection indicator says Supabase connected.

2. Select RBCET for the shared RBCET / RBMI Central Library, RBCP for Pharmacy, RBCN for Nursing, or RBGN for its separate library. RBMI is a campus in the shared library, not a fourth issue desk.

3. The single network administrator is rahul.r@rbmi.in and manages all four libraries. Nidhi Dewal, nidhi@rbmi.in, serves RBCET/RBMI. Rekha, rekha@rbmi.in, serves RBCP. Sheeba, sheeba@rbmi.in, serves RBCN. Raveena, raveena@rbmi.in, serves RBGN.

4. Use the sidebar modules. On a phone open the navigation menu. Refresh data to see a colleague's recent changes. Sign out before leaving a shared computer.

Notes and limits

A yellow Sample data banner means an isolated browser demonstration with fictional records. Live staff must use the Supabase-connected deployment.

Passwords are supplied privately to the owner. A patron card number is not a sign-in password; student import does not create authentication accounts.

02 / Accounts and role permissions

Permissions are enforced by the database using memberships and designated librarian assignments. Editable user profile metadata is never used to grant access.

How to use this section

1. Admin manages all libraries, policies, accounts, assignments and audit records. The system permits one network administrator.

2. A librarian manages the assigned library's catalogue, patrons, circulation, fines, availability, acquisitions, authorities and serials. Each library has a distinct primary librarian.

3. Assistant operates circulation and holds. Accountant records payments and waivers. Auditor reads records and reports. Staff access is limited to assigned libraries.

4. Patron sign-ins must be linked to a patron UUID. They see the catalogue and their own loans, holds and fines, and can renew their own loans or manage their own holds.

5. Admin opens Settings > Accounts to create staff, update non-admin email addresses or send recovery messages. Sensitive actions require the administrator's current password.

6. To provision a patron sign-in, create its email/password identity in Supabase Auth, then assign patron role with the linked patron UUID in Institution access.

Notes and limits

The create-account form supports librarian, assistant, accountant and auditor roles. Library records and Auth users are separate entities.

Retained sample accounts ending in .invalid cannot receive recovery emails. Replace them with real individual accounts before operational use.

03 / Change or recover a password

Email recovery is available to every role, including the administrator, through Supabase Auth and the configured SMTP service.

How to use this section

1. When signed in, select Change my password. Enter the current password, a different new password and matching confirmation.

2. When unable to sign in, choose Forgot password? Recover by email. Enter the registered address and check inbox and spam folders.

3. Open the newest recovery link. Continue securely if prompted, then enter and save a matching new password. Successful recovery signs you out; sign in with the new password.

4. Use 12 to 128 characters, including uppercase, lowercase, a number and a symbol. Choose a unique password for each person.

5. Admin may reset another assigned account in Settings > Institution access after verifying the person's identity and entering the administrator password. Share the temporary password privately and ask the person to change it.

6. For an expired, incomplete or previously used link, request a new email. Ask admin to correct a wrong registered email.

Notes and limits

Staff cannot reset the administrator; the administrator recovers through their own registered email.

Old passwords cannot be read from Supabase. Passwords are excluded from audit records, GitHub and public source downloads. An initial-password file becomes stale after users change their passwords.

04 / Overview and the daily desk routine

Overview shows physical-copy and bibliographic-title totals, active and overdue loans, posted outstanding fines and collection status for the selected library.

How to use this section

1. At opening verify the library, designated librarian and availability panel. Record any librarian absence for the relevant date.

2. Review Loans needing attention and waiting holds. Use Issue a book, or open Circulation for return and renewal.

3. Verify patron identity and book barcode for every transaction. Review any rejection before repeating it.

4. Refresh after colleagues' changes. At closing export required reports, inspect audit entries and follow the backup procedure.

Notes and limits

Outstanding fines shows assessed unpaid balances, not projected charges on every overdue active loan.

The application refreshes after its own changes. It does not subscribe to automatic real-time updates.

05 / Librarian availability and fine-free days

An unavailable date is a full-day exception for one library. The database records the date, reason, operator and change time. Partial-day closures are not modeled.

How to use this section

1. In Overview or Settings > Librarian availability, verify the selected library and librarian name.

2. Select the date, set availability off, enter a reason such as Librarian on approved leave, and save.

3. Confirm the history entry and the No fine badge for a closed current day. Correct an erroneous closure by restoring availability with a reason.

4. The designated librarian can edit today or a future date up to one year ahead. Only admin can correct historical dates.

5. Fine calculation skips unavailable dates only for this library. A due date that falls on a closed day moves to the next eligible open date.

6. Example: at INR 2 per eligible overdue day, three weekdays with one recorded unavailable day produce INR 4 before grace and cap. A date that is both a holiday and unavailable is excluded once.

Notes and limits

An unmarked date follows the normal calendar. The system does not infer absence from failed logins or a librarian not signing in.

Historical calendar corrections do not rewrite an already assessed fine. Review the original return and use an authorized audited waiver where appropriate.

06 / Patrons and student profiles

Patrons are borrowing memberships, separate from Auth users. All supplied RBCET student files are assigned to the shared RBCET/RBMI library.

How to use this section

1. Open Patrons and search by name, code, category or department. Source card number is the patron code; roll number is used when no card number was supplied.

2. Expand View profile to inspect supplied roll/card numbers, institute, course, branch, year, semester, admission date, mobile, email, gender and birth date where present.

3. Add patron requires unique code, name, category and active status. Email, department/class and expiry are optional.

4. To edit a membership, retain its existing code. The core editor changes name, email, category, department, active flag and expiry; uploaded additional profile fields remain preserved.

5. Deactivate or expire memberships when borrowing must stop. Preserve their loan history. Export CSV only for authorized institutional work.

Notes and limits

Source blanks remain blank. No birth dates, contact information or expiry dates are invented.

Importing students does not create login accounts. Student profiles and source archives are private and never appear in anonymous OPAC responses.

07 / Bulk uploads and the supplied Excel import

The supplied workbooks were processed through a controlled administrative import. Regular application Bulk upload uses UTF-8 CSV, with preview and confirmation before a transactional batch.

How to use this section

1. Choose the correct library, open Catalogue or Patrons > Bulk upload, and download its template.

2. Books require title, author, accession and barcode. Optional fields include ISBN, publisher, subject, classification, year, shelf and price. One row creates one physical copy.

3. Student CSV requires code and name; optional email, category, department, active and expires. Additional profile columns can be preserved through student imports.

4. Export Excel as UTF-8 CSV without converting identifiers into scientific notation or losing leading zeros. Use no more than 1,000 rows and 5 MB per batch.

5. Review errors and the first eight rows, confirm the checkbox and import. A failed row rolls the entire batch back.

6. Existing patron codes update core memberships. Duplicate book barcodes/accessions reject the batch. Review updates carefully; blank expiry clears it and blank active defaults to true.

7. The owner-provided import prepares 278 students and 26,071 book copies. One book row missing a barcode or title is held in the private review report. Original workbooks remain unchanged.

Notes and limits

Protected archives retain source hashes, sheet names, row numbers and original fields. The BSCS sheet's incomplete header was mapped to the common 15-column book layout after inspection.

Missing authors use Unknown author. Missing or nonnumeric prices use zero operationally while retaining the original value in source details. Historical loans are not inferred from a stock list.

08 / Bibliographic records, copies and barcodes

A bibliographic record describes a title/edition. Each physical copy has its own accession, globally unique barcode, shelf, price and status.

How to use this section

1. Search Catalogue by title, author, ISBN, classification, barcode or accession. Use Show more to reveal further matching cards.

2. Add title requires title and author. Enter ISBN, publisher, subject, classification/DDC and publication year when known.

3. Open View copies > Add physical copy. Enter unique accession and barcode, shelf and nonnegative price.

4. Use Edit title to correct descriptive fields. Use Status with a reason for available, lost, damaged, withdrawn or missing copies. Loaned copies must first be returned.

5. Use Label to print a Code 128 barcode. Scan-test labels before applying them to books.

6. For imported copies, Source details shows original item type, collection, accession date, pages, publication place and notes when supplied.

Notes and limits

The provided workbook has barcodes but no separate accession column, so its barcode also serves as the accession.

The controlled import groups consistent descriptions into 9,668 bibliographic records. Regular CSV import reuses a matching non-empty ISBN; inspect editions before using that simpler path.

09 / MARC book migration

Catalogue > MARC migration supports UTF-8 ISO2709 .mrc and MARCXML .xml files, supplementing CSV and the controlled Excel import.

How to use this section

1. Prepare a UTF-8 file up to 10 MB and 1,000 physical copies. Convert MARC-8 through your cataloguing tools first.

2. Map 245 title, 100/110/700 author, 020 ISBN, 260/264 publisher/year, 650 subject and 082 classification.

3. Items use 952$p barcode, 952$i accession or barcode fallback, 952$a/$b library, 952$c shelf and 952$g price.

4. Select the file, review its preview and confirm the checkbox. Item library must match the selected workspace; RBMI is an alias for shared RBCET.

5. Import and inspect copy counts and identifiers. Invalid records, duplicate barcodes or cross-library items reject the whole batch.

Notes and limits

This is targeted MARC import, not a complete MARC editor, MARC-8 converter or lossless MARC export service.

Invalid record lengths, directories and terminators are rejected. XML document types and entities are prohibited.

10 / Issue, return and renewal

Issues store the library code/name, operator UUID, timestamp, barcode/accession, title and patron code snapshots. A book must return to its issuing library.

How to use this section

1. Choose the correct library and click Issue a book. Verify the explicit library code, scan barcode or accession and enter patron code.

2. Save. The server checks copy availability, membership/expiry, loan limits, overdue loans, balance threshold and FIFO hold priority. Due dates follow the library calendar.

3. Open Circulation and locate the active loan by book, barcode or patron. Verify the library column and due date.

4. Use Renew only when permitted. Overdue loans, exhausted renewals, waiting holds, inactive memberships and excessive balances block renewal.

5. Use Return in the same library. The server rejects another issuing library code even for admin. Return, fine posting and copy availability commit together.

6. Inspect the return notice and assessed fine. In Returned loans choose Receipt and print its library, patron, book and fine details.

Notes and limits

Each loan stores its policy values. Later policy edits apply to new loans; existing issues retain their recorded rules.

Do not repeat a successful return. The database rejects returns on a loan already closed.

11 / Holds and reservation priority

Holds reserve a bibliographic title within a library, using a FIFO waiting queue ordered by creation time and identifier.

How to use this section

1. Choose Holds > Place hold, or Place hold in a title's copy drawer. Select a title and enter an active, unexpired patron code.

2. Review queue position, placement time and status. The same patron cannot have duplicate waiting holds on the same title.

3. Issue the next available copy to the first waiting patron. The server blocks queue bypass and marks the hold fulfilled when issued.

4. Cancel unwanted waiting holds. Staff manage the library queue; patrons manage only their own holds.

Notes and limits

A waiting hold blocks renewals of that title.

Pickup-expiry timers, automatic notification emails and inter-library transfers are not implemented. Use the approved institutional contact procedure.

12 / Fine rules, payments and waivers

Fines are assessed on return, counting eligible dates after the due date through the return date under the issuing library calendar.

How to use this section

1. Admin opens Settings > Circulation policies to set loan days, maximum loans, renewals, daily fine, grace days, cap and blocking balance by category.

2. Add holidays in Settings. Weekend exclusion follows the saved policy. Holidays and explicit librarian-unavailable dates are excluded.

3. Subtract grace from eligible overdue days, floor at zero, multiply by daily rate and apply the cap. INR 2 times three eligible days minus one grace day equals INR 4.

4. Open Fines and inspect amount, paid, waived and remaining balance. Record Payment with a positive amount and receipt/reference.

5. Use Waiver with an authorized reason. Payment and waiver totals cannot exceed the assessed amount. Export and inspect the audit trail.

Notes and limits

Grace is counted in eligible days, not ordinary calendar days. Calendar changes do not recalculate posted historical charges.

Lost/damaged status does not automatically generate replacement-cost charges. Follow the institution's separate approval process.

13 / Acquisitions, budgets, funds and baskets

Acquisitions track vendors, orders and full receipt of copies. Budget/fund and vendor-basket controls implement the targeted workflows from the supplied Koha manuals.

How to use this section

1. Add a vendor and the bibliographic title to purchase.

2. Create a dated budget and allocate funds with unique codes. Total fund allocations cannot exceed the budget.

3. Create a vendor basket if grouping purchases. Order with vendor, title, invoice, quantity and unit price; optionally select fund and basket.

4. For a funded order, the budget must be current, basket vendor must match and committed plus received value must fit the fund.

5. Receive an ordered acquisition once, supplying accession/barcode prefixes. Receipt creates the full ordered quantity atomically.

6. Review order status, receipt dates and fund commitments. Export orders for accounts reconciliation.

Notes and limits

Unfunded orders are permitted; operationally select the approved fund when budget control is required.

Partial receipt, EDI, foreign-currency conversion, accounting ledgers and hierarchical subfunds are outside this release.

14 / Authority control

Authorities standardize personal names, corporate names and subjects and link them to bibliographic titles in the same library.

How to use this section

1. Open Authorities and create a personal, corporate or subject heading with preferred text and optional variants/notes.

2. Review spelling and punctuation to avoid duplicate standardized headings.

3. Use Link title to associate the heading with a local title. Cross-library associations are rejected.

4. Inspect linked titles while maintaining a heading and use available edit controls for corrections.

Notes and limits

This is a local heading/link workflow.

Full authority MARC editing, external harvesting and automatic global heading replacement are not implemented.

15 / Private and public reading lists

Lists collect titles for study or display. Private lists belong to their owner; public lists are visible to authenticated users in the same library.

How to use this section

1. Open Lists and create a name with private or public visibility.
2. Choose titles to add and the offered sort option. Only the owner can add or remove entries.
3. Review list contents and remove unwanted titles.
4. Use Holds separately to reserve a title for borrowing.

Notes and limits

Public list visibility is library-scoped and authenticated; it is separate from the anonymous OPAC cart.

Lists do not reserve physical copies and do not bypass circulation rules.

16 / Serial subscriptions and receiving

Serials manage journal subscriptions, expected issues and arrival states, separately from ordinary physical-copy circulation.

How to use this section

1. Create a subscription with title, frequency, first issue date and predicted issue count.

2. Select weekly, monthly, quarterly or yearly. Generate at most 120 issues per subscription.

3. Review expected dates/numbers. An anchor date of 31 January predicts 28 February and then 31 March.

4. Mark issues arrived, late or missing. Arrived issues store the receiving date.

5. Use statuses and expected dates for manual follow-up and reporting.

Notes and limits

Predictions do not send vendor claims automatically.

Irregular calendars, automated claims, routing and automatic circulating-item creation are outside this release. Create ordinary catalogue copies separately if needed.

17 / Advanced catalogue and item search

Search combines text terms with physical-copy condition and recorded checkout history in the selected library.

How to use this section

1. Choose title, author, ISBN, subject, classification, barcode, accession or shelf as the search field.

2. Enter one or two terms. AND requires both, OR accepts either and NOT excludes the second term in the selected field.

3. Filter by available, loaned, lost, damaged, missing or withdrawn status.

4. Use Checkout count at most and Last checkout on/before for low-use analysis. Never-issued copies have no last date and remain eligible for the date filter.

5. Review the first 100 rows or export all matches to CSV.

Notes and limits

This is a simple Boolean filter, not a full search-engine query language.

A stock spreadsheet does not contain historical circulation. Imported copies start at zero recorded checkouts until issued in this application.

18 / Public catalogue and temporary cart

Open Browse public catalogue or https://rbmi-ilms.vercel.app/?opac=1 without a staff account.

How to use this section

1. Select a library and search title, author, ISBN or subject.

2. Review publisher/year, classification, accession/shelf, status and available-copy counts.

3. Use Show more for additional matches. Add titles to the temporary cart and print the selection/catalogue page.

4. Return through Staff / reader sign in. Linked patrons manage their own loans and holds only after authentication.

Notes and limits

Anonymous data contains library names, bibliographic records and limited holdings. It excludes patron details, accounts, prices, fines and audit history.

Cart selections are temporary and reset on refresh. Purchase suggestions, reviews, tags, ratings and anonymous holds are outside this release.

19 / Stock verification

Stock verification compares scanned barcodes with a fixed collection snapshot. One open session is allowed per library.

How to use this section

1. Open Stock verification and create a named session for the selected library.
2. Scan copies against the session. Repeated scans do not add duplicate expected records.
3. Review seen and unseen items, especially copies that were already on loan at session start.
4. Investigate discrepancies before closing. Confirm closure and inspect resulting copy statuses.
5. Export findings for approval. Correct later copy status through Catalogue with a reason.

Notes and limits

The expected inventory is fixed at session creation; reconcile later acquisitions and circulation separately.

Do not classify every unscanned item as permanently lost without physical and circulation checks.

20 / Reports and saved definitions

Reports provide CSV exports and a safe dataset/date report builder. Arbitrary SQL is never accepted from the browser.

How to use this section

1. Export Loans, Catalogue, Patrons, Fines, Payments or Acquisitions as available in Reports.

2. Inspect Most borrowed titles for recorded checkout totals, including returned issues.

3. Create a named report for loans, copies, patrons, fines, acquisitions or serial_issues, with optional date bounds.

4. Run it, inspect the first 100 rows and download CSV for all results.

5. Dates apply to loan issue dates, acquisition order dates and serial expected dates. Copies, patrons and fines are current snapshots and ignore date bounds.

Notes and limits

Staff reporting is restricted to assigned libraries. Patron accounts cannot run staff reports.

Report scheduling, emailed outputs and Koha's arbitrary SQL editor are not implemented. Protect downloaded data.

21 / Administration and audit trail

The network administrator manages the four libraries. Authorization is checked on each database/API request.

How to use this section

1. Select the correct library before changing policies, holidays or assignments in Settings.

2. Create staff using real email addresses and unique temporary passwords. Creating a librarian also assigns that library's primary librarian.

3. To replace a librarian, use Assign librarian with the existing Auth UUID and display name. The former primary librarian membership is removed from that library.

4. Assign assistant, accountant, auditor or linked patron memberships in Institution access. Another network admin cannot be created here.

5. Open Audit trail, search by action/actor/details, expand stored details and export authorized records.

6. Review availability changes, account operations, imports, returns and fine waivers in particular.

Notes and limits

The administrator email is fixed in the account UI. An administrator-identity change requires deliberate trusted database maintenance.

Audit records have no user-facing deletion. Passwords are not stored in audit details.

22 / Developer installation and deployment

The source repository is drrahulrastogi/rbmi-ilms. The UI and account endpoints run on Vercel; shared records and Auth run on Supabase.

How to use this section

1. Install Node.js 20.16 or newer, run npm ci, npm test and npm run build. npm run dev starts local development. Without both Supabase variables it uses fictional browser-local records.

2. For a NEW Supabase project run database/install-new-supabase.sql once, containing all eight migrations and the policy/institution seed. Never run the creation installer over an existing library.

3. Disable public signup, keep email/password login enabled, create rahul.r@rbmi.in in Auth and run scripts/bootstrap-admin.sql with its UUID.

4. Set VITE_SUPABASE_URL and VITE_SUPABASE_ANON_KEY to the public URL/key. Set SUPABASE_SERVICE_ROLE_KEY only as a sensitive server-side Vercel variable; never prefix it with VITE_.

5. Set Auth Site URL to the HTTPS application, allow /?recovery=1, configure SMTP and require 12-character passwords with lower/upper letters, numbers and symbols.

6. Deploy the repository root to the intended Vercel Vite project. vercel.json runs tests before the build. The current release uses authenticated CLI deployment; GitHub source hosting alone does not imply automatic Vercel deployment.

7. For an existing installation, back up first and apply only missing migrations in order. Never recreate the original schema over live data.

Notes and limits

PostgreSQL provides transactional circulation and unique identifiers. Supabase supplies managed database/Auth and Vercel serves the UI and server endpoints.

Free plans have quotas and provider-controlled pause/retention rules. Check dashboards before scaling. No paid upgrade is enabled by this release.

23 / Backup, restore and private handover

The public source package includes schema and fictional seeds, not live students, staff passwords or original private workbooks.

How to use this section

1. Use an authorized database dump or trusted SQL export for all app tables, source archives and audit records. Store backups in a protected off-site location.

2. Inventory the Supabase/Vercel projects, environment variable names and SMTP setup. Keep secret values in an approved password manager.

3. Plan Supabase Auth identity backup/migration separately through supported tools. A patron CSV is not an Auth backup.

4. Before upgrades/imports take a verified backup and retain original Excel files and reconciliation reports.

5. Restore into a separate project and verify counts, memberships, loans, fines and recovery settings before trusting a restore procedure.

6. Treat initial-password CSVs as private handover files that become outdated after password changes; use recovery instead of assuming old passwords remain valid.

Notes and limits

The pre-upgrade database backup is stored privately in the workspace. A full disaster-recovery restore drill has not been performed for this release.

Clearing browser storage removes local demonstration records, not live Supabase data.

24 / Troubleshooting and Koha coverage limits

The supplied Koha documentation informed the acquisition, advanced/item search, authority, migration, circulation, lists, OPAC, patron, report and serial workflows. This application implements the targeted workflows, not complete Koha compatibility.

How to use this section

1. Wrong password: use the registered React-library email and email recovery. A NetBeans/Derby account or patron number is a different identity.

2. No membership: ask admin to assign the correct library/role. A replaced librarian may retain an Auth identity without access to the old library.

3. Issue rejected: check library code, copy status, active membership, expiry, loan limits, overdue loans, balance threshold and hold priority.

4. Fine seems wrong: inspect saved policy, eligible dates, holidays, availability, grace and cap. Already-posted charges require an authorized waiver when correcting past dates.

5. No email: inspect spam, use the latest request, verify the address, SMTP limits and Auth redirect settings. The owner confirmed administrator recovery-email delivery during setup.

6. Import fails: inspect required columns, duplicates and invalid numbers. A failing batch rolls back.

7. Slow catalogue: narrow the search and use Show more. Holdings load through secure server pages. Very large audit histories may require additional pagination.

Notes and limits

Outside scope: offline circulation, RFID/SIP2, EDI, SSO, federated discovery, complex serial schedules, automated claims/hold emails, purchase suggestions, reviews/tags, partial receipts and full MARC editing/export.

Same-library returns intentionally prevent inter-library returns. Report errors without sharing passwords, private keys or student exports.